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Credit Card Fraud Detection via Intelligent Sampling and Self-supervised Learning

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The significant increase in credit card transactions can be attributed to the rapid growth of online shopping and digital payments, particularly during the COVID-19 pandemic. To safeguard cardholders, e-commerce companies, and financial institutions, the implementation of an effective and real-time fraud detection method using modern artificial intelligence techniques is imperative. However, the development of machine-learning-based approaches for fraud detection faces challenges such as inadequate transaction representation, noise labels, and data imbalance. Additionally, practical considerations like dynamic thresholds, concept drift, and verification latency need to be appropriately addressed. In this study, we designed a fraud detection method that accurately extracts a series of spatial and temporal representative features to precisely describe credit card transactions. Furthermore, several auxiliary self-supervised objectives were developed to model cardholders' behavior sequences. By employing intelligent sampling strategies, potential noise labels were eliminated, thereby reducing the level of data imbalance. The developed method encompasses various innovative functions that cater to practical usage requirements. We applied this method to two real-world datasets, and the results indicated a higher F1 score compared to the most commonly used online fraud detection methods.

CCS Concepts: • **Computing methodologies → Artificial intelligence**; **Knowledge representation and reasoning**

Additional Key Words and Phrases: Self-supervised learning, feature engineering, intelligent sampling, discriminative representation, credit card fraud detection

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1 INTRODUCTION

The COVID-19 pandemic forced customers to shop online, which caused an increase in the number of fraudulent transactions, highlighting the importance of fraud detection methods. E-commerce shopping in the United States grew by 14.2% in 2021, and consumption through e-commerce websites has skyrocketed because of the availability of increasingly diverse methods of mobile payment. Losses from online fraud amounted to US$392 million in 2021, representing a significant increase of 59.3% compared to the losses in 2020. Compared with offline transactions, the details of online transactions are easier to obtain, which enables more complete and comprehensive data to be collected for analyzing customer behaviors. Therefore, studies have explored the effectiveness of novel deep learning techniques [57, 62], such as autoencoders and **self-supervised learning (SSL)** [60], in detecting and preventing fraudulent activities. Leveraging advanced artificial intelligence methods enables the attainment of more precise and comprehensive online fraud detection, thereby safeguarding cardholders, e-commerce companies, and financial institutions.

Credit card fraud detection system can be significantly improved by addressing various challenges, including insufficient representation, noise labels, and data imbalance. The information insufficiency of customers' behavior sequences and stores' consumption records arises from the limited store trading records. Enhancing the representativeness of these data is crucial for improving the effectiveness of credit card fraud detection. This can be achieved through various approaches, such as data enrichment, feature engineering, and data augmentation. Data imbalance and noise labels are major problems that affect the effectiveness of a credit card fraud detection system. In the real world, the number of genuine transactions is considerably higher than that of fraudulent transactions. This imbalanced data distribution results in two disadvantages: (1) models cannot learn the representation of frauds because of the limited data on fraudulent transactions and (2) models cannot identify useful characteristics from a large quantity of genuine but uninformative data. Moreover, two different types of noise labels will significantly reduce the effectiveness of the detection system's learning process. First, a transaction will only be reported as fraudulent by investigators or the cardholder after a certain period of time has passed. Second, credit card fraud may be concealed within everyday small-value transactions or regular consumption behaviors. Both of these noise data scenarios are challenges in the learning process for imbalanced data classification.

Some additional requirements, such as a dynamic threshold and handling concept drift [14, 59], must be taken into account to enhance the accuracy and practicality of the proposed credit card fraud detection system. After generating a list of suspected fraudulent transactions, investigators initiate phone calls to potential victims. The maximum number of calls within a specified period is restricted due to limited human resources, and this limitation is referred to as the threshold. The daily allocation of phone calls is designed to ensure both the workload of the manual recheck process and the accurate and efficient identification of fraud. Hence, a dynamic threshold is crucial to meet these requirements. The concept drift problem arises because customers' consumption habits and fraudsters' strategies change rapidly. For instance, some customers spend more money during the Christmas season. A sudden surge in spending might lead to prediction errors for a fraud detection system. To cope with the rapid evolution of fraud detection systems, fraudsters change their fraud strategy to avoid detection. Therefore, the concept drift problem must be considered to maintain the performance of a fraud detection system.

To address the aforementioned problems, in this study, a feature extraction method based on financial knowledge [23] was developed to detect different credit card fraud situations. In this method, various representations based on the transaction amount, temporal correlation, and spatial information are designed for better describing complicated user behaviors. Moreover, several novel SSL tasks are adopted to extract discriminative sequential behavior representations from raw transactions. To the best of our knowledge, we are the first to apply SSL techniques in the field of financial fraud detection. Problems related to noise labels and imbalanced data are alleviated using an intelligent sampling strategy [61]. The representative embedding of each transaction is obtained, the distance between them is calculated, and a fixed proportion of negative data is ignored. Practical problems such as a dynamic threshold and concept drift are solved efficiently and effectively in the proposed method. Experiments conducted on two real-world datasets demonstrated the superiority of the proposed method over the selected baseline methods. Additionally, we elaborate on the relationship between the dynamic threshold design and performance evaluation. The key contributions of this study are summarized as follows:

— A representative and discriminative feature extraction method based on expert financial knowledge was developed to describe various credit card fraud detection situations.
— The noise label and imbalanced data problems were alleviated using the proposed intelligent sampling strategy.
— Several SSL tasks, guided by the logic of consecutive transactions, were integrated into a deep neural network to extract distinctive sequential behavioral representations from raw data.
— Practical problems such as a dynamic threshold and concept drift were addressed to enable the application of the developed method to a commercial bank.
— The proposed method considerably outperformed different baseline methods regarding fraud detection by using real-word credit card transaction datasets.

The remainder of this article is organized as follows. Section 2 provides a comprehensive review of the relevant literature. Section 3 introduces the developed credit card fraud detection framework. Section 4 provides the problem definition and describes the adopted intelligent sampling strategy and SSL framework for user behavior sequences. Section 5 presents the experimental results for a confidential real-world dataset. Finally, Section 6 provides the conclusions of this study and recommendations for future research.

2 RELATED WORKS

In this chapter, four fields related to this article are introduced: online fraud detection, imbalanced data, noise label, and self-supervised learning.

2.1 Online Fraud Detection System

Fraud detection has been widely discussed due to the substantial losses caused by fraudulent transactions. Supervised learning algorithms, such as logistic regression [46] and decision trees [21], are commonly employed to build predictive models that classify transactions as fraudulent or legitimate based on labeled training data. With the rapid advancement of artificial intelligence, deep learning techniques have emerged as a powerful approach for tackling online fraud detection [19, 34]. However, in certain deep learning methods, where each transaction is processed independently, this approach cannot precisely model the user's continuous transactional behavior. To overcome this problem, several methods have been developed to obtain user behavior patterns by inputting user behavior sequences as additional information into deep learning models. Bernardo et al. [7] proposed the use of **Gated Recurrent Units (GRU)** in building a credit card fraud detection model. Their research introduced efficient batch training and streaming inference techniques, eliminating the necessity for extensive feature engineering and enabling real-time detection of financial fraud. Reference [37] introduced a novel user behavior sequence pretrain model utilizing self-supervised learning techniques at various levels of granularity. The proposed model aims to capture and extract meaningful patterns from user behavioral data. Reference [38] proposed a local intention calibrated tree-LSTM combined with an attention model for analyzing user behavioral data.

Recently, graph-based approaches have gained significant attention and adoption [44]. Li et al. [35] introduced the **Life-streaming Fraud Detection (LIFE)** model, which leverages a tripartite heterogeneous graph neural network for live-streaming fraud detection. This study was the first to apply **Graph Neural Networks (GNNs)** to live-streaming fraud detection scenarios. Additionally, [68] proposed a multi-view attributed heterogeneous information network for detecting financial defaulters on online payment service platforms. Liu et al. [40] developed a label-balanced sampler and a neighborhood sampler to address class imbalance challenges in graph-based fraud detection. Reference [10] proposed a method that utilizes graph neural networks and attention mechanisms to learn spatial and temporal information for effective fraud detection. These studies demonstrate the growing interest and advancements in leveraging graph-based techniques for fraud detection applications. While many studies have demonstrated the performance improvement achieved by incorporating sequential data and graph data into deep learning models for fraud detection, the representations of users and items are often constrained by the available data collected. Therefore, we have devised several discriminative features leveraging financial domain knowledge and employed data augmentation techniques to enhance the representativeness of the original data.

2.2 Machine Learning with Imbalanced Data and Noise Label

Imbalanced data is a common problem where there is a significant disparity in the number of samples among different classes. This issue is prevalent in various applications, including medical diagnosis, credit card fraud detection, and others. This problem poses several drawbacks to machine learning algorithms as they tend to bias their predictions towards the majority class [29]. To tackle this issue, common methods include resampling techniques [52] such as oversampling, which augments the representation of minority samples, and undersampling, which diminishes the influence of majority samples. Oversampling methods, such as **Synthetic Minority Oversampling Technique (SMOTE)** [8] and its variants, have widely applied in various fields. In Reference [13], an encoder/ decoder module was proposed based on SMOTE to deal with imbalanced problem in image field. Undersampling techniques, such as Tomek Links [42] and Edited Nearest Neighbors [71], involves removing majority samples to enable the classifier to make more equitable decisions. Another common approach is cost-sensitive learning, where different misclassification costs are assigned to each class to increase the accuracy of the model [58]. Through this mechanism, the classifier can prioritize the minority class and improve the results accordingly. Ensemble methods have also demonstrated good performance in addressing the issue of imbalanced data [64]. Techniques such as bagging [45] and boosting [16] are effective in mitigating the impact of imbalanced data.

In recent years, with the advancement of computational power, deep learning algorithms have also been widely applied to address the issue of imbalanced data [18]. Reference [17] proposed an anomaly detection approach based on GAN for multivariate time series. They designed a novel training strategy for the generator to adjust weights based on reconstruction errors. Reference [25] solved imbalanced data distribution problem by proposing an edge pruning policy searching module. Reference [50] considered homophilic and heterophilic connections simultaneously to construct a graph-based fraud detector. Additionally, transfer learning is another technique used to deal with imbalanced data problem. Reference [55] applied transfer learning on a AdaBoost-CNN and successfully reduced the computational cost on several imbalanced datasets.

Noise labels present a significant challenge in numerous machine learning applications, since they denote instances in the training dataset that with incorrect labels. Such mislabeling may lead to inefficient learning process and a decrease in model performance. Numerous studies have focused on researching how to mitigate the impact of noise labels. There are four main machine learning approaches to dealing with noise labels: robust architecture methods, robust loss design methods, robust regularization methods and sample selection methods [54]. Robust architecture methods involve adding a noise adaptation layer or developing a specifically designed architecture [63]. The disadvantage of the former method is that it is not able to identify falsely labeled data, while the disadvantage of the latter method is its limited applicability to other different approaches or methods. Robust loss design includes robust loss functions [66] and loss adjustment techniques [4]. The core idea of these approaches is to develop different loss function and loss adjustment approaches that are less sensitive to misclassified instances or by adjusting the loss based on the likelihood of noise presence. Regarding robust regularization [51], weight decay, drop out, data augmentation, and batch normalization are four commonly used regularization techniques. These techniques not only help mitigate the problems caused by noise labels but also improve the generalization ability of the models. Sample selection methods [27, 53] involves multi-network training and multi-round learning. The core concept of these methods is to select true-labeled data by co-training or refinement techniques. Furthermore, recent advancements in deep learning have also addressed the challenge of noise labels. The contrastive learning framework has also been widely employed to address the challenges associated with noise label problems, primarily due to its excellent representation capabilities [32]. In this study, we integrate sampling approaches and deep learning methods to address the challenges of imbalanced data and noise label problems in constructing credit card fraud detection systems.

2.3 Self-supervised Learning

SSL is a type of unsupervised learning method in which the main goal is to learn a common feature expression for performing downstream tasks. According to Reference [39], SSL frameworks can be categorized into generative and contrastive SSL frameworks. In generative SSL, an encoder is trained, and the encoded input is fed into a decoder that must reconstruct the original input. In contrast to generative SSL that uses a decoder, in contrastive SSL, after encoder training, the similarity of differences between the explicit vector generated by the encoder and another explicit vector is determined. SSL has been applied to a wide range of applications, including computer vision [30] and the recommendation field [69]. These applications have utilized SSL for purposes such as data augmentation and discovering concealed patterns within sequential data. In Reference [70], a masking technique was used to obtain user attributes, items, sequences, and associations between items and user attributes. For capturing the hidden correlation between them, mutual information maximization [5] was conducted to obtain not only behavior patterns but also additional information on items, sequences, and user attributes.

Several studies in the field of self-supervised learning have relevance to our research, particularly in the context of anomaly detection within time series data. Reference [65] developed an adaptive memory network to enhance the generalization ability for anomaly detection on multivariate time series data. The method demonstrates superior performance compared to several state-of-the-art baselines on four public datasets. Zhang et al. [67] devised multiple self-supervised learning tasks and integrated them to obtain a better representation for detect anomaly patterns. The method in Reference [24] presented a multiresolution self-supervised discriminative network for anomaly detection. Reference [28] proposed a novel self-supervised contrastive loss to distinguish

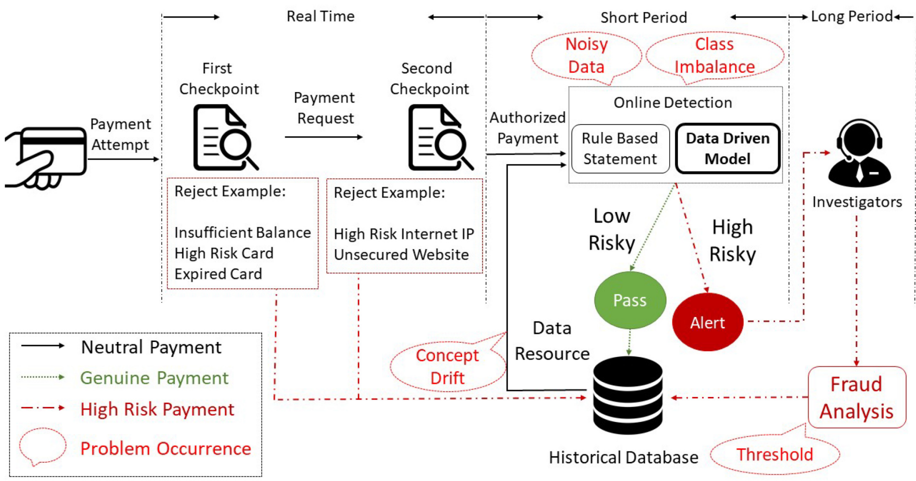

Fig. 1. Illustration of credit card payment flow.

pseudo negative time series and original time series. The above researches apply self-supervised learning techniques in multivariate time series data to conduct anomaly detection. What differentiates their research from ours is that the time series data they used is not financial in nature. In our approach, we took into consideration the characteristics of financial data, specifically the fact that an individual's consumption behavior should exhibit consistent patterns. Based on this characteristic, we designed self-supervised learning tasks to leverage the inherent patterns in the data.

In the financial field, numerous studies have been conducted on fraud detection including Reference [22]. Reference [11] proposed a group-aware approach based on graph neural network to detect money laundering activities. Liang et al. [36] devised a device-sharing network for insurance fraud detection. Reference [2] utilized generative adversarial networks and ensemble learning techniques to identify fraudulent activities in financial statements. However, self-supervised learning techniques are not widely utilized in the majority of research works focusing on financial fraud detection. The advantage of self-supervised learning lies in its ability to learn effective representations without relying on labeled data. This characteristic allows us to discover representations that capture the differences between fraudulent and normal samples, thus enabling us to distinguish fraudulent behaviors more effectively. Therefore, we have proposed several novel SSL tasks specifically designed to extract discriminative sequential behavior from raw transaction data. Our work is the first to introduce and apply SSL strategies in the field of financial fraud detection.

3 CREDIT CARD PAYMENT FLOW

The real-world credit card payment flow is described in this section [15]. Figure 1 illustrates the entire verification process from the time when a payment is made to the time when the payment is recorded in a historical database. The real-world credit card payment flow comprises four steps: (1) terminal checkpoint, (2) rule-based checkpoint, (3) fraud detection, and (4) fraud analysis.

3.1 Terminal Checkpoint

As soon as a transaction is generated, traditional security checks are performed on the payment request. The terminal, which is a device that receives credit card information after the consumption, plays a crucial role in the terminal checkpoint phase. It is capable of returning detailed error messages, including those related to insufficient balance and exceeded expenditure limits. The terminal has the capability to identify various issues related to credit cards, such as expiration, unidentified status, or even being flagged as fake or fraudulent in the database of the issuing bank. This allows the terminal to halt transactions promptly. In the terminal checkpoint phase, it is essential to report the result within a few milliseconds.

3.2 Rule-based Checkpoint

After verification through the terminal checkpoint, a payment request is subjected to the rule-based checkpoint phase. In this phase, the system analyzes the request according to several rules without using historical records. For instance, if a transaction is generated through a high-risk merchant or Internet protocol, then the payment request is denied for security reasons. The rules followed in the aforementioned phase are designed by experts or investigators, and they must be highly precise to avoid unnecessary manual analysis. The entire rule-based checkpoint phase must be performed in real time. Transactions that are verified through the rule-based checkpoint are considered to be authorized. These transactions are then subjected to fraud detection for further examination.

3.3 Fraud Detection and Analysis

Because of their rich hidden information, many fraudulent transactions remain unidentified after the aforementioned two phases. In the fraud detection phase, a score is assigned to an authorized payment. The higher this score is, the greater the likelihood of the transaction being fraudulent. Fraud detection models can be divided into two categories: rule-based models and data-driven models. This study focused on obtaining a better description of transaction representations for better detecting fraudulent transactions in the fraud detection stage.

The design of rule-based models is more complicated than that of the rules of the rule-based checkpoint and usually involves the use of transaction details. For example, if a transaction request appears at midnight, but the user has never conducted a transaction from the relevant store around this time, then the fraud score of the transaction is high. However, the concept drift problem can lead to the decrease of the performance of rule-based models, which forces experts or investigators to change the rules frequently. This process is inefficient and causes the wastage of human resources. Data-driven models include a transaction classifier that is trained to identify fraudulent transactions by determining the probability of authorized payments by using various algorithms. These models extract consumption patterns automatically in a short period by considering multiple features simultaneously, which makes them highly flexible. This study aimed to improve the performance of a data-driven model, including its feature engineering, data augmentation, and updating steps.

Investigators, including experts, must analyze fraud patterns and design rules for the rule-based checkpoint and rule-based models. Experts analyze the records in the alert pool and ask specialists to contact cardholders, if necessary. This phase might require a few days to a few weeks because of human-related factors, such as nonresponse from the user. After the fraud status of a payment is confirmed, the payment is recorded into a database for future use.

4 PROPOSED METHOD

In this section, feature engineering methods and a self-supervised pretraining technique are proposed to enhance the representation of training data. An intelligent sampling strategy and a dynamic threshold strategy are further adopted to construct a credit card fraud detection model. Figure 2 illustrates system overview of the proposed method, which includes several key

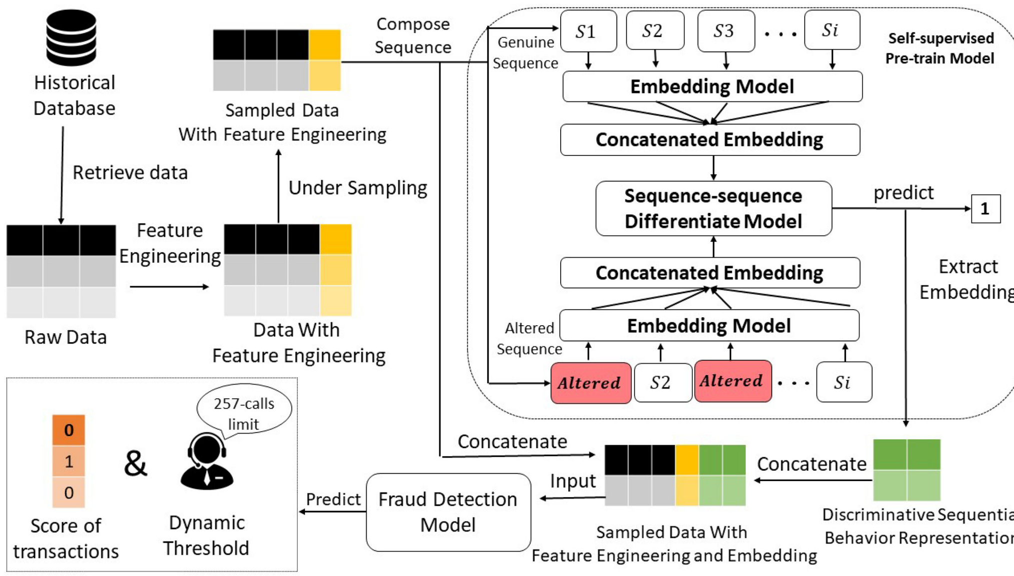

Fig. 2. System overview of the proposed credit card fraud detection method.

contributions of our approach. The notations used are defined and summarized into Table 1. The feature engineering method is introduced in this study to extract representative features based on expert financial knowledge. By applying the intelligent sampling method, the noisy label and class imbalance problems can be alleviated, leading to improved model performance and robustness in credit card fraud detection. Several self-supervised learning tasks are proposed to discover hidden patterns among transactions. Finally, practical issues will be addressed in Section 4.5, such as the dynamic threshold. This section explores the trade-off between investigator workload and detection accuracy.

4.1 Feature Engineering Methods Based on Financial Knowledge

In addition to sequence embedding learned directly by deep neural networks, developing a numerical method to represent sequence behavior is crucial for learning effectiveness. To achieve this goal, our model incorporates transaction sequence information to enhance system performance. There are two ways to include sequential information into our model. One is directly using sequential modeling method such as recurrent neural network or long short-term memory network on raw data. The other is obtain sequential information from feature engineering method and put them into a deep neural network to train an end to end model. The former method can extract useful information from raw data but may need lots of training data and time to construct a model. The latter approach can shorten the training time of the model, since some features have already been calculated. However, the input is limited to the engineered features, which may lack diversity in the features. In this section, we introduce the feature engineering methods we have designed based on financial knowledge.

Inspired by References [12, 26], our feature engineering methods can be divided into four categories: transaction amount-based representation, temporal correlation-based representation, spatial information-based representation and immediate high-risk alert feature. Most of features originated from feature engineering methods are calculated by using the user's past transactions within two months. In other words, in addition to the transaction information itself, a transaction

Table 1. Notation Table for the Proposed Method

Tags	Symbol	Definition
Transaction amount	a_i	ith transaction amount
	m	medium of transaction
Temporal correlation	t_i	ith transaction time (in second)
Spatial information	C_i	country where ith transaction generated
	c_i	city where ith transaction generated
Sampling	v_n^M	nth majority sample
	v_n^m	nth minority sample
	$d(M, m)$	Euclidean distance between two sequences
	$D(M)$	accumulation of $d(M, m)$
SSL model	$\mathcal{A}$	altered sequence
	$\mathcal{S}$	genuine sequence
	m^e	transaction embedding from altered sequence
	s^e	transaction embedding from genuine sequence
	d	embedding dimension
	$\mathcal{A}_e$,	concatenated embedding of altered sequence
	$\mathcal{S}_e$	concatenated embedding of genuine sequence
	$y^{\mathcal{A}}$	prediction of altered sequence
	$y^{\mathcal{S}}$	prediction of genuine sequence
Training Objective	$\mathcal{T}$	whole transaction with feature engineering
	t	transaction
	t^*	transaction with embedding
	$\mathcal{T}^*$	whole transaction with embedding added
Dynamic Threshold	T_v	threshold for validation set
	T	threshold for testing set
	q	quota on suspected fraud for testing
	q_i	quota on day i in the testing period
	I	total days of the testing period
	$FDNN$	fraud detection model
	P	transactions predicted as fraud
	$R1$	fraud ratio
	$R2$	fraud ratio after normalization

data also includes several preceding transactions, such as the amount, time, and location of the previous transaction. This additional information helps the model better understand the background and context of the transaction, thereby predicting whether it is a fraudulent transaction more accurately.

4.1.1 Transaction Amount-based Representation. The transaction amount is the most important factor in a transaction, because it represents the consumption habit of a user. For instance, some customers subscribe to streaming services monthly, which involves a series of fixed-amount transactions, while some customers use credit cards only when performing high-value transactions. Since a single transaction offers limited information, feature engineering leverages previous transaction records associated with the same credit card or the card owner as a data source to achieve more comprehensive representations. The notations $A_1, A_2, \ldots, A_{16}$ represent different transaction amount features. The current transaction amount is denoted as a_i, and the previous transaction records within the past 2 months are represented as months $a_1, a_2, \ldots, a_{i-1}$, where $i \in \mathbb{N}$.

— Amount difference (A_1): If the difference between the current transaction amount and the previous transaction amount is large, then the current transaction is suspicious. This difference A_1 is calculated as follows:

$$A_1 = |a_i - a_{i-1}|, \tag{1}$$

where || represent absolute value.

— Aggregated amount (A_2): The aggregated amount A_2 is determined as follows to enhance the representation of A_1:

$$A_2 = \frac{\sum_{n=1}^{i} a_n}{i}. \tag{2}$$

— Medium Calculation (A_3): The medium of previous transactions m and the current transaction amount a_i are used to calculate A_3 as follows:

$$A_3 = (|a_i - m|)/m. \tag{3}$$

This feature helps the model recognize pattern deviations.

— Number of identical-amount transactions (A_4): The number of identical-amount transactions in a day is determined, because certain fraudsters attempt to avoid detection by the rule-based model by making multiple transactions of the same small amount.

— Standard deviation (A_5): The standard deviation of the current transaction from previous transactions is calculated to obtain additional information.

— Interval for all previous transactions (A_6): This categorical feature is calculated to determine whether the current transaction amount is within 1 standard deviation from the amount of previous transactions.

— Average daily amount of the transactions made by customers with a merchant (A_7, A_8): The average amount spent by customers in the transactions of a day on each merchant is calculated. The fraud detection model determines the difference between the current transaction amount and the aforementioned amount for a merchant. The standard deviation of the transaction amount of each merchant is also calculated.

— Calculations for the merchant ($A_9, \ldots, A_{14}$): Single credit card and card-holder's: (1) average transaction amount in the same merchant; (2) the standard deviation of transaction amount in the same merchant; (3) accumulated amount in the same merchant are calculated.

— Interval for previous transactions with a certain merchant (A_{15}, A_{16}): This feature is calculated to determine whether the current transaction amount for a merchant is within 2 standard deviation of the amount spent for the merchant. If the current amount is not within this standard deviation, then the transaction might be fraudulent, because it represents abnormal customer behavior.

4.1.2 Temporal Correlation-based Representation. In fraud detection models, acquiring temporal information from previous transactions is crucial, as it reflects the consumption habits and patterns of a user. For instance, some people visit the same store every Friday night to buy daily necessities, which results in the generation of weekly transactions. Therefore, an irregular transaction occurring at a different merchant might to be fraudulent. To obtain more detailed information, the time unit used in this section is seconds. The notation $T_1, T_2, \ldots, T_9$ represent temporal correlation features. The current transaction timestamp is defined as t_i, and the timestamps for previous transaction amounts within the past 2 months are denoted as $t_1, t_2, \ldots, t_{i-1}$, where $i \in \mathbb{N}$.

— Time difference (T_1): Fraud may be indicated by the following scenarios: (1) a credit card that has been unused for a long time generates an authorized transaction, and (2) an unusual series of transactions is generated within a few seconds. Therefore, the time difference T_1 between transactions is calculated as follows:

$$T_1 = |t_i - t_{i-1}|. \tag{4}$$

- Angle (T_2): An equation including the timestamp and transaction amount is used to calculate the "angle" between the current and previous transactions. When a_i increases or D_{t_i} decreases, an abnormal transaction is indicated. The arc tangent of the ratio of the aforementioned factors is used to calculate the angle to prevent the angle from increasing markedly:

$$angle = \arctan(a_i/D_{t_i}). \tag{5}$$

- Time transformation (T_3, T_4): We transfer timestamp into actual date and calculate the result with sine or cosine formula by using record of hour t_i. The formula is as follows:

$$sin\left(\frac{t_i}{24} \cdot 2\pi\right), \tag{6}$$

and

$$cos\left(\frac{t_i}{24} \cdot 2\pi\right). \tag{7}$$

- Midnight transaction analysis (T_5, T_6): Fraudsters are often more active at midnight, which is between 12 a.m. and 6 a.m. Therefore, the proportion of midnight transactions is calculated. With regard to the categorical feature, whether the transaction is generated during midnight hours is determined.
- Von Mises Distribution (T_7): We referred to Reference [6] to check whether the transaction fell within a von Mises distribution for extracting the periodic transaction behavior.
- Accumulated transactions with a certain time (T_8): If numerous transactions are generated by a credit card in a short period, then the transactions might be fraudulent. Therefore, the numbers of transactions within 1, 10, 60, 3,600, and 7,200 s are counted.
- Time-based interval (T_9): We examine whether the time interval between t_i and t_{i-1} falls within various predefined periods to extract temporal information for the current transaction.

4.1.3 Spatial Information-based Representation. The location of a transaction serves as an indicator of a consumer's spending behavior. When a transaction originates from a different country or city than the user's usual locations, it is considered abnormal and requires additional analysis. The notations $L_1, L_2, \ldots, L_6$ represent location-based features. The merchant for the current transaction is located in country C_i and city c_i, and the previous records within the past 2 months are donated as $C_1, C_2, \ldots, C_{i-1}$ and $c_1, c_2, \ldots, c_{i-1}$, where $i \in \mathbb{N}$.

- Country and city statistics (L_1, L_2): the number of C_i or c_i appears in $C_1, \ldots, C_{i-1}$ or $c_1, \ldots, c_{i-1}$ is counted. If the current transaction is conducted from a different country or city than usual, then the transaction might be a fraudulent one.
- Number of countries (L_3): The number of countries in which a credit card is used is determined.
- Proportion of transactions in a country (L_4): The proportions of transactions for C_i and C_1 to C_{i-1} are calculated as supplementary representations for L_1.
- Country Comparison (L_5): For categorical feature engineering, whether C_i is the same as C_{i-1} is checked to determine whether the consumption location has changed.
- Country Existence (L_6): Whether C_i appears in previous transactions is checked to search for abnormal patterns.

4.1.4 Immediate High-risk Alert. This type of feature originates from information provided by domain experts, who informed us that fraud tends to recur at specific stores or with certain cardholders. Therefore, fraud rate statistics for a specific object can provide additional fraud patterns for a fraud detection model. The notations $F_1, F2, F3, F_4$ represent the fraud rates for a specific merchant, cardholder, credit card, and transaction type, respectively, within the previous 2 months or 1 week. Records on the transaction type records indicate how the transactions were generated. Transaction types include point-of-sale, canceled, and pretrade transactions.

The feature engineering enables a fraud detection model to differentiate between the representations of fraudulent and genuine transactions. For instance, if the country from which the current transaction is conducted is different from the countries in which previous transactions have been conducted, then the current transaction might be fraudulent. Such abnormal consumption patterns can more easily extracted by a model after feature engineering. According to our later experiment, the aforementioned features considerably affect a model's fraud detection performance.

4.2 Embedding-based Intelligent Sampling

To address the class imbalance and noise label problems, embedding-based intelligent sampling is adopted to reduce the quantity of negative labeled data for improving model efficiency and retaining informative samples. There are several steps in the embedding-based intelligent sampling. First, a credit card fraud detection model is pretrained based on a deep neural network, where the input comprises the features of each transaction, and the output is a binary classification indicating whether the transaction is fraudulent or not. After training the model, each transaction is fed into it again to obtain its hidden representation from the middle layer, which captures the transaction's latent factors. This embedding enables us to represent each transaction in a lower-dimensional space. Then, for each non-fraudulent transaction, the nearest fraudulent transactions will be selected and calculate the average distance between these fraudulent transactions and the non-fraudulent transaction. Non-fraudulent transactions that are close to fraudulent transactions will be removed. We do this because we suspect these transactions, although not labeled as fraudulent, might represent noise-labeled data due to their resemblance to fraudulent transactions in the embedding space. The entire process is illustrated in Figure 3. In contrast to the random sampling method, where some samples with crucial information might be overlooked, the embedding-based intelligent sampling method utilizes a heuristic rule to select informative samples. Instances of the majority class are then eliminated based on their Euclidean distance to their neighbors. A longer distance between negative and positive samples enables easier detection by a model. Assume that each sample contains N features, and that v_n^M and v_n^m represent the embedding of the majority sample M and the minority sample m, respectively, with $1 \leq n \leq N$. The Euclidean distance between the majority and minority samples (i.e., $d(M, m)$) can be calculated as follows:

$$d(M, m) = \sqrt{\sum_{n=1}^{N} \left(v_n^M - v_n^m\right)^2}. \tag{8}$$

The majority samples are genuine transactions (negative data), whereas the minority sample are fraudulent transactions (positive data).

Noise label data refer to fraudulent transactions that are labeled as genuine transactions during model training. Because the embedding of noise label data might be similar to that of positive data, the Euclidean distances between noise label and positive data are calculated. Subsequently, a fixed proportion of negative data is eliminated, which facilitates the elimination of noise label data and reduces the level of data imbalance, because the proportion of negative data is decreased. To prevent information loss while eliminating negative samples, multiple neighbors must be considered.

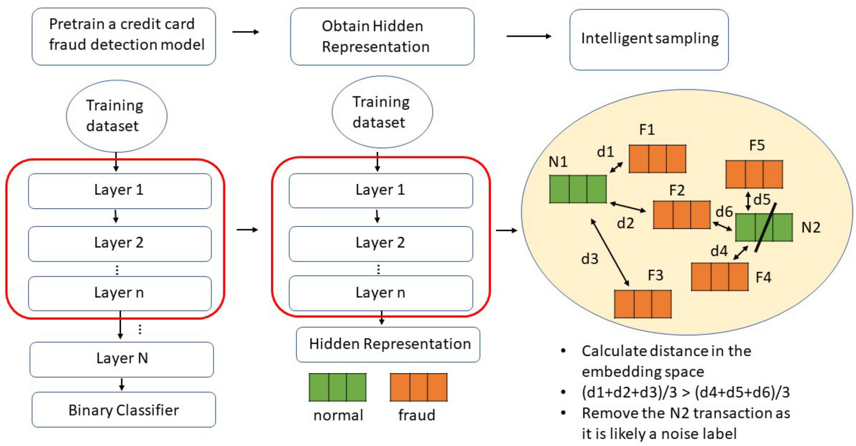

Fig. 3. Process of the embedding-based intelligent sampling strategy.

The distance from M to K neighbors $D(M)$ is expressed as follows:

$$D(M) = \frac{\sum_{i=1}^{K} d(M, m_i)}{K}. \tag{9}$$

4.3 Self-supervised Learning for Transaction Sequence Representation

Studies have indicated that an appropriate sequence length of transactions can ensure that sufficient information is retained while maintaining system efficiency. In addition to extracting sequential information through feature engineering methods, the short length of user transaction sequences (with an average of only four transactions per credit card and most credit cards having only one transaction record) makes it challenging to capture the relationships between different sequences solely through feature engineering. Therefore, we also utilize SSL techniques to help us gather more information. SSL, chosen for its ability to define custom learning objectives during training without the need for pre-labeled data, enables us to uncover hidden patterns and capture interdependencies among different sequences. This approach not only overcomes the limitations of short sequential data length but also automates the feature learning process, facilitating the extraction of intrinsic information to improve the efficiency of credit card fraud detection.

Discriminative sequential behavior representations can be used for effectively modeling the correlation between customers and items. Inspired by Reference [70], in which SSL was conducted on sequential data, we attempted to determine the intrinsic correlations between sequential transactions to enhance transaction data representations. The mutual information maximization method is adopted in the proposed approach, because it can effectively capture the correlation between data originating from the same input. Additionally, many domain experts in the field of credit card fraud detection have provided valuable insights to us. Based on their suggestions, we have designed various auxiliary tasks that leverage the logic of consecutive transactions to extract representative information. These auxiliary tasks fall into the following categories.

— **Consistency in Transaction Amounts.** Transaction amounts often strongly reflect a user's shopping habits, and the daily expenses within the same individual should generally remain consistent. There are two common fraudulent patterns related to transaction amounts: one involves a sudden large transaction, and the other comprises numerous very small transactions. For instance, if an individual who typically spends 10 dollars on daily meals suddenly makes a 20,000 dollars luxury purchase, then it could potentially be a fraudulent transaction. Additionally, the price range of items sold by a store typically falls within certain limits; for instance, convenience stores typically do not sell items costing tens of thousands of dollars. Hence, we employ masking techniques for store, product, and amount information to conduct contrastive learning. This approach enhances the representation of transaction sequence.

— **Continuity of Spatial Information.** Apart from transaction amounts, information about transaction locations is also crucial for detecting fraudulent behavior. The consecutive locations where a user makes transactions reflect their current position. For users who don't frequently travel abroad, most of their transactions should occur within the same country. Therefore, individuals whose transaction sequences span multiple countries are highly likely to be fraudulent activities. The locations of consecutive transactions are usually related. For example, if a previous transaction occurs at a physical store in Tokyo, and another physical transaction takes place in New York two hours later, then it may be indicative of potential fraud. Drawing from the insights of domain experts regarding patterns of fraudulent behavior, we have designed various auxiliary tasks related to transaction locations. We perform perturbation on the location information within the transaction sequence. The self-supervised technique learns whether a sequence of transaction locations is reasonable. These tasks are aimed at enhancing the representation of transaction sequences.

— **Reasonableness of Temporal Correlation.** Continuing with the discussion on fraudulent patterns related to transaction times, based on our data analysis, transactions occurring between midnight (0:00) and 6 a.m. represent the highest proportion of fraudulent activity. The timing of transactions is closely correlated with the location of transactions and is influenced by the time zone of the consumer. If someone suddenly places an order for food during non-meal hours or makes a significant purchase in the middle of the night, then there is a higher likelihood of it being a fraudulent transaction. During late-night hours, most transactions are typically online. If there is an in-person transaction during this time, then it may raise suspicion of fraud. We have also formulated these fraudulent behaviors into various auxiliary tasks. Through techniques like masking and reordering, we manipulate the transaction sequence to assess whether a transaction is fraudulent.

— **Correlation among Distinct Sub-Sequences within the Same Individual.** The sub-sequences highly reflect an individual's shopping habits and financial status. Therefore, we assert that the various sub-sequences of an individual should demonstrate consistent transaction characteristics. If an individual belongs to the middle-class, then most of his sub-sequences would typically involve transactions with similar amounts. However, if there is suddenly a substantial transaction amount within a particular sub-sequence, then it is highly likely that the transactions within that sub-sequence are fraudulent. By observing patterns of fraudulent activities, we gain some clues about fraudulent behavior. However, listing all fraudulent behaviors one by one would be impractical due to their diversity. Therefore, we segment an individual's transactions, breaking a single sequence into many sub-sequences. Subsequently, we employ contrastive learning on sub-sequences from different individuals. This auxiliary task helps the machine autonomously learn how to distinguish between fraudulent and non-fraudulent behaviors.

— **Connections Between Two Distinct transaction sequences.** Based on the credit card transaction data we have access to, consumers typically hold more than one credit card. They decide on the usage of each card based on the card's reward points or cashback. For example, a user might have three credit cards: one for booking international flights and hotels, another for everyday dining expenses, and a third for purchasing luxury items. Our auxiliary task's objective is also to determine if different sequences belong to the same individual. However, compared to the previous task, this one is more challenging, because the differences in spending across various consumption categories can be significant. Nevertheless, we believe that there are still specific patterns in an individual's credit card usage habits, even across different spending categories. Therefore, we aim to use this approach to identify anomalous behaviors that deviate from a user's typical spending habits.

The aforementioned data augmentation approaches enable a fraud detection model to distinguish between genuine and altered sequences. The quantity of altered data in a sequence affects the learning performance of the model. An inadequate or excessive quantity of altered data increases the difficulty of the model identifying hidden representations in sequences. Our self-supervised learning approach distinguishes itself from other studies by considering the characteristics of financial data, particularly the consistent patterns expected in an individual's consumption behavior. This unique approach is our novelty and differentiates us from other self-supervised learning research.

Self-supervised learning encompasses various approaches, and in this study, sequences are modified through techniques such as masking, perturbation, reordering, segmenting, and more. Genuine sequences are then compared to altered sequences to uncover hidden representations. An overview of the entire process is illustrated in Figure 2. Assume that $\mathcal{A}$ and $\mathcal{S}$ donate a altered sequence and genuine sequence of i transactions, respectively (i.e., $\mathcal{S} = \{s_1, \ldots, s_i\}$ and $\mathcal{A} = \{m_1, \ldots, m_i\}$). Each transaction of these two sequences is embedded into a low-dimensional vector (s^e and m^e) by using a deep neural network to extract discriminative information. Subsequently, each sequence embedding is separately aggregated into a longer dimensional embedding in the following manner:

$$\mathcal{S}_e = \left[s_1^e : s_2^e : \ldots : s_i^e\right], \tag{10}$$

$$\mathcal{A}_e = \left[m_1^e : m_2^e : \ldots : m_i^e\right], \tag{11}$$

where the concatenated embeddings of the genuine and altered sequences are denoted by $\mathcal{S}_e$ and $\mathcal{A}_e$, respectively. These concatenated embeddings have $i \cdot d$ dimensions. Informative embeddings can be obtained through the aforementioned concatenation process, and these embeddings prevent representations from being too scattered and can maintain the transaction order. After the concatenation process, a deep neural network is used to differentiate between genuine and altered sequences. The adopted prediction equation is $|y^{\mathcal{S}} - y^{\mathcal{A}}|$, where $y^{\mathcal{S}}$ and $y^{\mathcal{A}}$ represent the predictions for a genuine sequence and a altered sequence, respectively. The loss function of the sequence differentiation model is expressed as follows:

$$\mathcal{L} = -\sum_i y_i \cdot log\left(\left|y_i^{\mathcal{S}} - y_i^{\mathcal{A}}\right|\right). \tag{12}$$

After the completion of model training, embeddings are extracted as the additional information for each transaction and as the input data of the fraud detection model. Algorithm 1 describes the process of the adopted SSL model. The term N denotes the total number of pairs of genuine and altered sequences, and DIFFMODEL represents the deep neural network. Self-supervised learning has been rarely utilized in financial fraud detection research. To the best of our knowledge, we are the first to apply this technique to credit card fraud detection and extract discriminative sequential behavior from raw transaction data to enhance the performance of our fraud detection system.

4.4 Training Objective

The proposed method comprises four stages: feature engineering, embedding-based intelligent sampling, self-supervised learning, and credit card fraud detection. In the first three stages, representative and discriminative features are extracted to model various credit card fraud detection scenarios. In this section, our fraud detection model is described. In addition to the features generated through the feature engineering process and the sequence embeddings obtained from SSL training, we also include raw transaction features in our credit card fraud detection system. In total, there are approximately 100 different features. We denote the whole transactions as $\mathcal{T}$. For example, a transaction t can be associated with 10 previous transactions conducted with the same credit card within 60 days, where $t \in \mathcal{T}$. These 10 transactions can be combined into a sequence, whose embedding can be obtained using a well-trained, self-supervised pretraining model. This embedding is concatenated with t to represent the additional features, which is denoted by t^*, and the whole transactions with embedding concatenated is donated as $\mathcal{T}^*$.

ALGORITHM 1: Self-supervised Learning Process for Transaction Sequence Representation

Input: N **pairs of** $\mathcal{S}$ **and** $\mathcal{A}$

Output: Loss summed by Equation (12)

```
for each (S, A) ∈ N pairs of sequences do
    for each s ∈ S do
        s^e ← EmbModel(s)
    end for
    for each m ∈ A do
        m^e ← EmbModel(m)
    end for
    S^e ← Concatenate([s^e_1 : ... : s^e_i])
    A^e ← Concatenate([m^e_1 : ... : m^e_i])
    y^S ← DiffModel(S^e)
    y^A ← DiffModel(A^e)
    L ← |y^S − y^A|
end for
ℒ ← Sum(log(|L|) · y)
```

Fraud detection, which is achieved using a **fraud detection neural network (*FDNN*)**, is the final stage of the proposed method. An FDNN contains a deep neural network whose input is $\mathcal{T}^*$. Transactions are input into the binary classifier of this network, and the network outputs whether the transactions are fraudulent. The loss function of the credit card fraud detection task is binary cross-entropy, which is expressed as follows:

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N} y_i \cdot log(\hat{y}_i) + (1 - y_i) \cdot log(1 - \hat{y}_i), \tag{13}$$

where y_i is the true label (0 or 1), and $\hat{y}_i$ is the prediction score by the model.

4.5 Dynamic Threshold Design

To confirm the authenticity of fraudulent transactions, investigators must analyze the transactions and contact with the cardholders. However, due to limited human resources, there is a constraint on the number of calls that can be made within a specific period, and this limit is referred to as the threshold. In addition, the threshold will be automatically adjusted based on the estimated fraud rate. When the estimated fraud rate is lower and human resources are tighter, the threshold will be adjusted higher to avoid missing any potential fraudulent transactions. Conversely, when the fraud rate is higher and human resources are more abundant, the threshold will automatically be adjusted lower to capture as many fraudulent transactions as possible. Therefore, we propose a strategy called dynamic threshold to enable efficient fraud investigation. The dynamic threshold is designed to establish a daily quota for suspicious transactions while maintaining a reasonable level of model performance, without surpassing the maximum limit that credit card companies can handle. Thus, $(q - P)$ must be minimized, whereas precision and recall must be maximized, where q is a quota for suspected fraud, and P denotes the number of transactions predicted as being fraudulent. For each $FDNN$ model, suspected fraudulent transactions are determined using Equations (14) and (15) when not adopting and adopting a dynamic threshold, respectively:

$$FDNN(trans) \begin{cases} \geq 0.5: & \text{Suspected fraud transaction,} \\ < 0.5: & \text{Normal transaction,} \end{cases} \tag{14}$$

$$FDNN(trans) \begin{cases} \geq T: & \text{Suspected fraud transaction,} \\ < T: & \text{Normal transaction.} \end{cases} \tag{15}$$

To satisfy the requirement for credit card fraud detection, the proposed dynamic threshold method involves several key steps, as follows:

4.5.1 Initial Threshold based on Validation Data (DTI). A dynamic threshold is aimed at controlling the number of suspected fraud transactions predicted each day. The initial threshold T_{init} is acquired from validation data for testing period I. The validation data are the data collected in the 14-day gap between the collection of the training and testing data. When a dynamic threshold is used in the adopted $FDNN$ to predict the transactions in the validation set, the network predicts that the number of suspected fraudulent transactions P_i is close to the average number of transactions that a bank can process in a week (q_{avg}, where the amount is denoted by q). Equation (16), in which $T_v = T_{init}$ and $q_{avg} = \frac{q}{I}$, presents the formula for the number of suspected fraudulent transactions in a day:

$$P_i = Count(FDNN(trans_{valid} \geq T_v)) \approx q_{avg}. \tag{16}$$

However, it is inadequate for predicting fraud across all testing ranges by relying solely on the initial threshold determined using the validation set. This is because a static initial threshold may not accurately account for the varying number of fraudulent transactions that should be allocated each day. In addition, the threshold based on the validation set may not be well reflected in the testing set, which may lead to excessive prediction errors in some test intervals. Therefore, we propose the methods described in the following text to improve the effectiveness of our dynamic threshold method.

4.5.2 Adjusting Prediction Threshold (DTII). Another key aspect of dynamic thresholding is that it allows the dynamic adjustment of the threshold for the next day according to the actual predicted number of frauds on the current day, which reduces the prediction error in the final testing interval. One method for adjusting the threshold involves obtaining the number of frauds P_i predicted by the testing model on the previous day and allocating the remaining quota q_{remain} evenly over the remaining testing time.

The testing threshold at the beginning T_{init} is also obtained using the validation set. The number of frauds predicted by the adopted FDNN is expected to match the average number of frauds processed in a week (q_{avg}). Next, the quota for the remaining testing time (q_{remain}) is determined from the actual number of fraudulent transactions predicted on the first testing day, and the threshold for the next testing day is calculated. This step is repeated until testing is complete. By updating the daily threshold in the aforementioned manner, the error between the predicted number of frauds P and the tolerable quota q can be effectively decreased during testing. The step is shown in Algorithm 2.

ALGORITHM 2: Adjusting prediction threshold (DTII) Strategy

do
  $T_i \leftarrow T_v$
  $P_i \leftarrow Count(F(trans_{test\ i} \geq T_i))$
  $q_{remain} \leftarrow (q - P_i)$
  $q_{avg} \leftarrow q_{remain}/(I - i)$
  $T_{i+1} \leftarrow T'_v \;\; s.t.\ Count(FDNN(trans_{valid} \geq T'_v)) \approx q_{avg}$
while testing data is not end

4.5.3 Dynamic Threshold based on Validation Statistics (DTIII). The error in fraud prediction can be effectively reduced by adjusting the prediction threshold (DTII); however, the allocation of daily frauds is still based on q_{avg}, and the available information is not fully utilized. A method is proposed to acquire the rate of frauds P_{ratio} predicted by the daily validation set on a weekly basis and to assign the rate of frauds that testing transactions should be assigned to each day. In this article, we propose two parameters ($R1$ and $R2$) for calculating the fraud rate for the validation set. These parameters are expressed as follows:

$$R1_i = \frac{P_i}{P_{total}}, \tag{17}$$

$$R2_i = normalize\left(\frac{P_i}{trans_i}\right). \tag{18}$$

The parameter R1 is the ratio of the number of frauds predicted per day (P_i) to the number of frauds predicted per week (P_{total}). The parameter R2 is the normalized predicted fraud volume for a day when considering the transaction volume ($trans_i$). After obtaining the number of frauds from the validation set, Algorithm 3 is used for adjusting the threshold.

ALGORITHM 3: Dynamic threshold based on validation statistics (DTIII) Strategy

do
  $q_i \leftarrow q * R_i$
  $T_{init} \leftarrow T'_v\ s.t.\ Count(FDNN(trans_{valid} \geq T'_v)) \approx q_i$
  $P_i \leftarrow Count(FDNN(trans_{test\ i} \geq T_{init}))$
  $q_{remain} \leftarrow (q - P_i)$
  $q_{avg} \leftarrow q_{remain}/(I - i)$
  $T_{i+1} \leftarrow T'_v \;\; s.t.\ Count(FDNN(trans_{valid} \geq T'_v)) \approx q_{avg}$
while testing data is not end

In this method, the initial fraud threshold T_{init} is determined according to the percentage of frauds in the validation set, and the subsequent threshold is then adjusted on the basis of the actual number of predicted frauds and the statistics. Moreover, in the aforementioned method, statistical information is used to allocate daily frauds more reasonably, rather than using the average quota directly, and robust experimental results are obtained using the method.

5 EXPERIMENT

The proposed model was evaluated on two datasets and compared with different baselines methods. Several series of experiments were conducted to test the effectiveness of the proposed method.

5.1 Dataset and Evaluation Metrics

We measured the performance of our model by using two datasets, which are described as follows:

(1) *European Credit Card dataset*: This dataset contains 284,807 transaction records of European credit card holders from September 2013. Of these records, only 492 are positive (fraudulent transactions). Most of the features are transformed through principle component analysis.
(2) *Dataset of a major commercial bank in Taiwan*: This dataset contains 133,688,582 transaction records from May 2020 to November 2020. Of these records, 57,571 are fraudulent transactions (i.e., 0.04% of the entire dataset). The dataset contains details such as location, merchant, transaction amount, and payment method.

To assess the effectiveness of the proposed model, three evaluation metrics were employed including recall, precision, and F1 score. Furthermore, we assess the effectiveness of the imbalanced training method using the **Precision-Recall curve (AUC_PR)** due to its ability to capture the performance of classifiers in imbalanced datasets. Our framework is implemented using Python 3.4 and Tensorflow 1.12 on an NVIDIA Tesla P100 with 3,584 cuda cores. The operating system is Ubuntu 16.04.

5.2 Baseline Methods

The proposed model was evaluated by comparing its performance with that of several baseline models. We categorized the adopted baseline methods into two groups. The first category consists of well-known classification and regression methods in credit card fraud detection, such as LSTM-CRF published in 2022 [20]. Additionally, there are numerous methods that further incorporate imbalanced training strategies, which are more closely related to our proposed method. These methods are categorized as the second group.

(1) Learning-based Classification and Regression Approaches
   - **Linear regression model.** This model examines the relationships between multiple independent and dependent variables [3].
   - **Random forest model.** This model consists with multiple decision trees for avoiding the overfitting problem and improving the prediction performance [47].
   - **XGBoost.** This model is a type of gradient-boosting decision tree model in which the original model is unchanged each time and a new function is used to correct the errors of the previous tree [1].
   - **Adaboost.** Adaboost is an iterative model, in which the samples misclassified by the previous classifier are used to train the next classifier, and a new weak classifier is added in each round of classification until a predetermined small error rate is reached [43].
   - **LightGBM.** This model uses the weak learner based on a decision tree to iterate and obtain the best results. This model can be optimized for effectively increasing the training speed and reducing the computational resource consumption [41].
   - **LSTM-CRF Model.** This model incorporates a long short-term memory network and a CRF layer, which enables the capturing of temporal information in transaction data [20].
   - **Proposed Fraud Detection Neural Network (FDNN) model.** This model uses a deep neural network to detect frauds.
(2) Fraud Detection with imbalanced Training Strategy
   - **Random under sampling (RUS).** This is the most naive and commonly used method for imbalanced data training. RUS can lead to faster training times, but because it randomly drops data, it may result in the loss of a significant amount of valuable information.
   - **One-Class Support Vector Machine (OC-SVM).** This is an unsupervised algorithm commonly used for anomaly detection. It relies on a single class of training data, typically

Table 2. Comparison of Different Learning-based Classification Methods

	Precision	Recall	f1-score
Linear Regression	0.93	0.57	0.71
Random Forest	0.82	0.63	0.71
XGBoost	0.73	0.61	0.66
AdaBoost	0.70	0.58	0.63
LightGBM	0.88	0.70	0.78
LSTM-CRF	0.88	0.76	0.80
FDNN (FR = 1.3‰)	**0.94**	0.71	0.81
FDNN (FR = 1.5‰)	0.86	**0.77**	0.81
FDNN (FR = 1.4‰)	0.92	0.75	**0.83**

representing normal samples, to learn a decision boundary. The goal is to capture the characteristics of the normal data and identify deviations from it as anomalies [49].

- **Synthetic Minority Over-sampling Technique (SMOTE).** This is an oversampling method used to address class imbalance in datasets. It aims to increase the representation of the minority class by synthesizing new samples [8].
- **Overlap-sensitive Margin (OSM).** This method is a classifier specifically designed to address imbalanced data. It combines the **k-nearest neighbor (k-NN)** algorithm with a modified fuzzy **support vector machine (SVM)** approach [31].
- **Tomeklinks.** This method measures the distance between majority and minority samples and then removes those that fall within the borderline, as they may be noise. After removal, it can aid the classifier in achieving better classification.
- **NB-Tomek.** This method is an undersampling technique specifically designed to address the issue of overlapping data points in imbalanced datasets. It aims to remove potential overlap between the minority and majority classes by selectively eliminating majority samples from the overlapping region [56].
- **Dynamic weighted entropy (DWE).** This method consists of two steps: (1) exclusion of a substantial number of majority samples and several outliers from the minority class, and (2) application of a non-linear classifier to differentiate the overlapping subset [33].

5.3 Comparison in Terms of Different Intelligent Sampling Strategies

We initially assessed our embedding-based intelligent sampling method using a publicly available dataset. Subsequently, we applied our SSL approach to the dataset with raw transaction sequences to conduct a more detailed comparison and evaluation of the performance of our proposed method.

Table 2 presents the results obtained with different methods for the Kaggle dataset. To illustrate various real-world scenarios, the criteria for identifying fraudulent transactions will be dynamically adjusted based on the workload of the human resources and the estimated **fraud rate (FR)**. When there is a large overall transaction volume, the relatively strict criterion is established. Conversely, when the workload of human resources is more abundant, the threshold is adjusted to be relatively loose to capture more fraudulent transactions. Changes in the threshold affect the values of precision and recall. Therefore, in this experiment, we explore the variations in precision, recall, and F1-score for three different estimated fraud rates to investigate their implications in practical applications. The results indicated that among the machine learning methods, LightGBM outperformed the other methods in terms of recall and F1 score. Although the precision of the linear regression model was higher than that of LightGBM, the recall of the linear regression model was

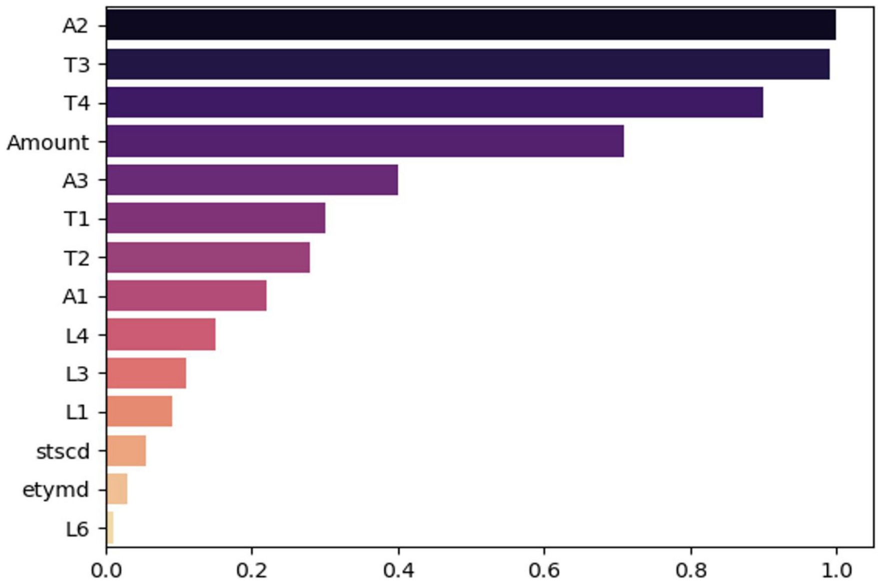

Fig. 4. Feature importance extracted from the LightGBM model.

only 0.57. These results indicated that the linear regression model predicted fewer positive labels than did LightGBM, which is undesirable. A suitable credit card fraud detection system should have high precision and recall. In terms of the deep learning methods, the LSTM-CRF model outperforms machine learning methods in all three evaluation metrics. We attribute this superiority to the model's architecture, which involves passing the user's transaction sequence through an LSTM layer to obtain hidden states. These hidden states act as input for the CRF layer, which generates results indicating the likelihood of each transaction being fraudulent. The LSTM-CRF model adopts a sequence-based approach, allowing it to effectively extract information from transaction sequences. FDNN model also outperformed all the adopted machine learning methods, which can be attributed to the fact that deep neural networks can capture more intrinsic information from data than machine learning models.

Regarding the comparison under different fraud rates, when the estimated fraud rate is lower and human resources are tight, higher precision and lower recall values are obtained. For example, at FR = 1.3‰, precision is 0.94, and recall is 0.71. Conversely, when the fraud rate is higher and human resources are abundant, higher recall and lower precision are observed. For instance, at FR = 1.5‰, precision is 0.86, and recall is 0.77. In the scenario with a moderate fraud rate FR = 1.4‰, the highest F1-score can reach 0.83. From the table, we can infer that our model can adjust the threshold based on real-world conditions, and the best performance under each metric surpasses that of LSTM-CRF. The results indicated that the proposed method effectively addressed the challenges of imbalanced data and noise labels, enabling the extraction of a substantial quantity of useful negative data and significantly improving model performance.

The feature importance figure (Figure 4) reveals that the initial representative features are primarily associated with transaction amount and temporal correlation. This suggests that transactions deviating from previous records in terms of amount or time are more likely to be fraudulent. While spatial information features are not as influential as the aforementioned ones, they still hold significant value in constructing credit card fraud detection systems. Additionally, three raw data features, namely, transaction amount (referred to as "Amount" in the figure), transaction type

Table 3. Comparison of Different Imbalanced Training Methods

	F1-score	AUC_PR
RUS	0.61	0.56
OC-SVM	0.67	0.57
SMOTE	0.64	0.54
OSM	0.69	0.59
Tomeklinks	0.71	0.66
NB-Tomek	0.66	0.56
DWE	0.73	0.63
FDNN	**0.81**	**0.71**

(referred to as "stscd" in the figure), and status code (referred to as "etymd" in the figure), also demonstrate significant importance. Overall, the figure highlights the substantial contribution of feature engineering methods in fraud detection, and suggests the potential for designing additional features related to transaction amount and temporal correlation to further enhance the model's capabilities.

Table 3 presents a comparison of the performance of our method with various imbalanced training methods. As shown in Table 3, the imbalanced training methods such as RUS, OCSVM, SMOTE, OSM, Tomeklinks, and NB-Tomek exhibit inferior performance compared to the DWE model. The superior performance of the DWE model can be attributed to its utilization of the "Divide-and-Conquer" strategy, which involves dividing the dataset into overlapping and non-overlapping subsets and classifying them separately. This approach enables the classifier to focus on different classes of samples, resulting in improved performance. Our proposed method outperforms the DWE method, indicating that our embedding-based intelligent sampling strategy effectively eliminates noise labels, reduces the level of data imbalance and achieves superior performance in credit card fraud detection.

5.4 Comparison in Terms of Different Dataset

The dataset of a major commercial bank in Taiwan was used for further evaluation. To avoid misjudgment by a fraud detection system, investigators from banks make phone calls only to those customers who might have been affected by fraudulent payment. However, because of the shortage of human resources, limited phone calls can be made in a certain time period. According to the information provided by the commercial bank, its investigators make approximately 5,000 phone calls per week. Therefore, one of our experiment settings was that only transactions with the highest fraud probability were considered fraudulent transactions. The term @5000 represents the total number of fraudulent transactions that the bank can confirm. To verify whether the proposed model is generalizable, we divided the training data by week into 10 rolling groups and implemented the model on the data for each week.

The objective of the experiment was to investigate the impact of noise labels and a decrease in the proportion of potentially negatively labeled data on the model performance. Six of the ten data groups were used as inputs for the proposed fraud detection model after the removal of different proportions of potentially negatively labeled data. The recall and average recall values were calculated, because our aim was to determine the detection rate of actual fraud. The results obtained for these settings were compared with those obtained for the settings without sampling (row 5 in Table 4).

In Table 4, r1–r6 indicate the rolling period. We compared our proposed method with several baseline approaches. The results presented in this table indicate that the elimination of a proportion

Table 4. Performance under the Elimination of Potentially Negatively Labeled Data at Different Proportions

recall@5000	r1	r2	r3	r4	r5	r6	Average
Random Forest	0.38	0.40	0.44	0.46	0.41	0.34	0.40
XGBoost	**0.42**	**0.42**	0.49	0.49	0.35	**0.35**	0.42
AdaBoost	0.37	0.36	0.39	0.39	0.27	0.27	0.34
LightGBM	0.34	0.31	0.44	0.43	0.22	0.27	0.33
FDNN W/O removal	0.38	0.31	0.48	0.48	0.55	0.32	0.42
FDNN with 0.0001% removal	0.41	0.32	0.34	0.49	0.54	0.29	0.43
FDNN with 0.001% removal	0.37	0.37	0.48	0.49	0.50	0.30	0.42
FDNN with 0.01% removal	0.40	0.38	0.48	0.51	**0.56**	0.29	**0.44**
FDNN with 0.1% removal	0.40	0.35	**0.50**	**0.52**	0.54	0.29	0.43

of potentially negatively labeled data enhanced model performance. However, the elimination of excessive data resulted in a decrease in recall, because useful data were deleted or the remaining data did not contain considerable noise. By contrast, the elimination of insufficient data resulted in noise label affecting the learning performance. According to our results, the elimination of 0.01% of the potentially negatively labeled data alleviated the class imbalance and noise label problems and yielded in the best performance (with 2% improvement).

5.5 Analysis on Concept Drift

In the model training process, due to the scarcity of positive data, we needed to determine an appropriate training period to gather a sufficient number of fraudulent transactions. Because of the concept drift problem, an excessively short or long training period worsens model performance. This phenomenon occurs because the behaviors of customers and fraudsters might change with time. The existence of insufficient positive data also reduces model performance. Therefore, the training period must be dynamically adjusted to acquire the best results. To examine the effect of the concept drift problem, an experiment was conducted using different training periods. Table 5 presented the results, where row 1 shows that all the positive data were collected from the previous rolling period.

The experimental results revealed that longer training periods benefited learning, because the proposed model could obtain additional information from positive labeled data. However, when using five or all rolling groups of positive data for training, recall decreased by 3%, which indicated that the concept drift problem eliminated the positive effect of a longer training period on model performance. The results indicated the negative effect of adopting excessive positive data for training. In addition, model performance was marginally worse when using one, two, or three rolling groups of data for training than when using five rolling groups of data for training. Thus, the model could not achieve ideal results when the training data were insufficient. The best results (3% higher than any other settings) were obtained when using four sets of rolling positive data.

5.6 Evaluating the Design of the Proposed Method

To confirm that all steps in the proposed method are essential in credit card fraud detection, an ablation experiment was conducted. Ten rolling groups of transaction data were used to check our model's stability and alleviate the concept drift problem. The descriptions of the models used in the ablation experiment are provided as follows:

- The **FDNN + MTSSL** model [48] employs multi-task self-supervised learning to enhance the representation of financial time series data for portfolio management.

Table 5. Impact of Different Training Periods for Concept Drift Analysis

Train periods	r6	r7	r8	r9	r10	Average
all	0.42	0.43	0.45	0.42	0.32	0.41
5	0.42	0.43	0.45	0.42	0.32	0.41
4	**0.43**	**0.44**	**0.46**	**0.42**	**0.33**	**0.42**
3	0.42	0.44	0.44	0.41	0.33	0.41
2	0.42	0.43	0.44	0.42	0.32	0.41
1	0.42	0.42	0.44	0.41	0.31	0.40

Table 6. Performance of the Proposed Credit Card Fraud Detection Method and Its Variants

model	r1	r2	r3	r4	r5	r6	r7	r8	r9	r10	Average
FDNN + MTSSL [48]	0.57	0.26	0.32	0.46	0.40	0.36	0.40	0.41	0.47	0.28	0.39
FDNN + S3 [70]	0.64	0.38	0.33	0.46	0.40	0.41	0.44	0.46	0.46	0.36	0.43
FDNN	0.60	0.36	0.31	0.50	0.42	0.41	0.42	0.45	0.40	0.32	0.42
FDNN + Period(4)	0.60	0.39	0.29	0.55	0.45	0.42	0.43	0.48	0.43	0.32	0.44
FDNN + SSL	0.64	0.34	0.32	0.48	0.39	0.42	0.44	**0.49**	**0.49**	0.36	0.44
FDNN + SSL + Period(4)	0.64	0.40	0.34	0.55	0.46	0.44	0.42	0.43	0.44	0.35	0.45
FDNN (with 0.01% removal) + SSL + Period(4)	**0.65**	**0.4**	**0.34**	**0.55**	**0.47**	**0.46**	**0.45**	0.49	0.43	**0.39**	**0.46**

— The **FDNN + S3** model followed Reference [70] to conduct self-supervised learning.
— The **FDNN** model is a fraud detection model based on deep neural networks.
— The **FDNN + Period(4)** model uses the positive labeled data of four training periods.
— The **FDNN + SSL** model uses differentiate model to discover the difference between the two sequences.
— The **FDNN + SSL + Period(4)** model uses the positive labeled data of four training periods to obtain additional fraud patterns.
— The **FDNN (with 0.01% removal) + SSL + Period(4)** model uses the proposed method, which includes the credit card fraud detection process designed in this study. This model obtains additional information through SSL, uses the positive labeled data of four training periods, and adopts the embedding-based intelligent sampling method to eliminate noise data.

Table 6 reveals that the proposed model outperformed the other adopted models. The FDNN + MTSSL model exhibited the lowest recall at only 0.39. This performance difference may be attributed to the fact that while MTSSL also conducts data augmentation on time series data, its downstream task is portfolio management. In contrast, our task is credit card fraud detection, where MTSSL may be more suited for representation learning related to portfolio allocation.

The FDNN + S3 model displayed better performance than the original FDNN. However, when the FDNN incorporated the mechanism of adjusting the training period, it outperformed even the S3 model. By introducing various methods into this model, including intelligent sampling and our designed SSL approach, the model's performance continued to improve. It achieved a 4% enhancement over the original FDNN, demonstrating that our proposed method can yield superior credit card fraud detection results.

5.7 Analysis on Dynamic Threshold

Our last experiment involved evaluating our model under a dynamic threshold. The threshold for each day was predicted in each rolling training period and compared with the corresponding number of actual frauds. The difference between the actual and predicted number of frauds was defined as the error. The error for each day was accumulated over a rolling period for evaluating

Table 7. Performance under Different Dynamic Threshold Strategies

	r1	r2	r3	r4	r5	r6	average
DTI	0.39	0.45	0.48	0.47	**0.55**	**0.38**	**0.45**
DTII	**0.40**	**0.46**	0.52	**0.48**	0.34	0.36	0.43
DTIII$R1$	0.39	0.42	0.52	0.45	0.31	0.36	0.41
DTIII$R2$	0.38	0.42	**0.52**	0.46	0.42	0.37	0.43

Table 8. Error Number under Different Dynamic Threshold Strategies

	r1	r2	r3	r4	r5	r6	Average
DTI	308	288	1266	813	49	851	595.83
DTII	93	41	193	158	28	84	99.50
DTIII$R1$	101	**20**	264	**112**	**5**	**11**	**85.41**
DTIII$R2$	**53**	139	**94**	129	104	218	122.93

our algorithm. A suitable fraud detection model should have high recall and low error. The methods used for dynamic thresholding in this study are described as follows:

- In the DTI method, the fixed average quantity is used as the standard for prediction.
- In the DTII method, the remaining fraud prediction quota for each day is considered to ensure the higher flexibility of the average quantity.
- In the DTIII $R1$ method, the ratio of frauds predicted each day is used.
- In the DTIII $R2$ method, the normalized predicted fraud volume is used.

Table 7 presents that the DTI method exhibited the best recall; however, it also exhibited the highest error (Table 8), which was 485% higher than the second-highest error. The aforementioned results can be explained by the fact that an excessive number of frauds are predicted in the aforementioned method, which considerably increases the number of errors. The DTII method exhibited a lower recall and error than did the DTI method. A dynamic threshold is more flexible than a fixed initial threshold, which is unable to respond to variation in the actual number of frauds. A trade-off existed between the DTIII $R1$ and DTIII $R2$ methods. The DTIII $R1$ method had the lowest error number (85.41) but also had the lowest recall. The DTIII $R2$ method exhibited higher recall but also had higher error than did the DTIII $R1$ method. In practice, the error is a more crucial factor than is the recall. When using the DTIII $R1$ method, only a few errors occurred in several rolling periods, which is useful in human resource allocation. The average training time for our best-performing configuration, which incorporates all the credit card fraud detection processes designed in this study, was approximately 20 h. Thanks to the pre-calculation and storage of feature engineering and self-supervised learning metadata, we achieved a remarkable speed per transaction in terms of testing time, which can meet the practical demand. Our method is now ready for commercial deployment and has demonstrated its effectiveness in assisting banks with credit card fraud detection.

6 CONCLUSION

In this study, a credit card fraud detection system was constructed to solve several challenges, including insufficient representation, noise labels and data imbalance. Practical issues such as concept drift and dynamic threshold were addressed in the developed system. We extracted representative features based on financial knowledge to describe various fraud detection situations and used SSL techniques to obtain discriminative sequential behavior representations. An intelligent sampling strategy was adopted to alleviate the noise label and reduce the level of data imbalance problems. Additional functions, such as a dynamic threshold and an adjustable training period, were designed to enable the developed system to meet real-world requirements. The experimental results revealed that the proposed method outperformed several baseline methods.

While the developed system achieved remarkable results, there is room for further improvement. First, more complex SSL tasks can be designed to obtain more detailed patterns of user behavior. Second, to protect the privacy of different entities in cross-industry alliances, which are becoming increasingly common under globalization, federated learning can be incorporated into the developed system [9]. This step would enhance the robustness and scope of the developed system and enable additional data on customers and stores to be obtained.

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